Version 7.4.7

PVsyst - Simulation report

Standalone system

Project: Novohjyoti Sinha

Variant: New simulation variant
Standalone system with batteries
System power: 2600 Wp
Agartala - India

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Project summary

Geographical Site	Situation		Project settings	
Agartala	Latitude	23.80 °N	Albedo	0.20
India	Longitude	91.27 °E		
	Altitude	18 m		
	Time zone	UTC+5.5		

Weather data
Agartala
Meteonorm 8.1 (1991-2012), Sat=100% - Synthetic

System summary

Standalone system — **Standalone system with batteries**

PV Field Orientation		Near Shadings	User's needs	
Fixed plane		Linear shadings : Fast (table)	Daily household consumers	
Tilt/Azimuth	30 / 0 °		Constant over the year	
			Average	8.6 kWh/Day

System information

PV Array		Battery pack	
Nb. of modules	10 units	Technology	Lead-acid, sealed, Gel
Pnom total	2600 Wp	Nb. of units	24 units
		Voltage	48 V
		Capacity	1002 Ah

Results summary

Useful energy from solar	3065.74 kWh/year	Specific production	1179 kWh/kWp/year	Perf. Ratio PR	72.48 %
Missing Energy	56.47 kWh/year	Available solar energy	3365.08 kWh/year	Solar Fraction SF	98.19 %
Excess (unused)	193.34 kWh/year				

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General parameters

Standalone system — **Standalone system with batteries**

PV Field Orientation

Orientation
Fixed plane
Tilt/Azimuth 30 / 0 °

Sheds configuration

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Near Shadings
Linear shadings : Fast (table)

User's needs
Daily household consumers
Constant over the year
Average 8.6 kWh/Day

PV Array Characteristics

PV module
Manufacturer Generic
Model OS260P
(Original PVsyst database)
Unit Nom. Power 260 Wp
Number of PV modules 10 units
Nominal (STC) 2600 Wp
Modules 5 string x 2 In series

At operating cond. (50° C)
Pmpp 2322 Wp
U mpp 55 V
I mpp 43 A

Battery
Manufacturer Generic
Model BAE Secura Block Solar 12 V 3 PVV 210
Technology Lead-acid, sealed, Gel
Nb. of units 6 in parallel x 4 in series
Discharging min. SOC 20.0 %
Stored energy 38.5 kWh

Battery Pack Characteristics
Voltage 48 V
Nominal Capacity 1002 Ah (C10)
Temperature Fixed 20 °C

Controller
Universal controller
Technology MPPT converter
Temp coeff. -5.0 mV/°C/Elem.

Converter
Maxi and EURO efficiencies 97.0 / 95.0 %

Battery Management control
Threshold commands as SOC calculation
Charging SOC = 0.90 / 0.75
approx. 52.2 / 49.5 V
Discharging SOC = 0.20 / 0.45
approx. 46.6 / 48.3 V

Total PV power
Nominal (STC) 2.60 kWp
Total 10 modules
Module area 16.3 m²
Cell area 14.6 m²

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 22 mΩ
Loss Fraction 1.5 % at STC

Serie Diode Loss
Voltage drop 0.7 V
Loss Fraction 1.1 % at STC

Module Quality Loss
Loss Fraction -0.8 %

Module mismatch losses
Loss Fraction 0.5 % at MPP

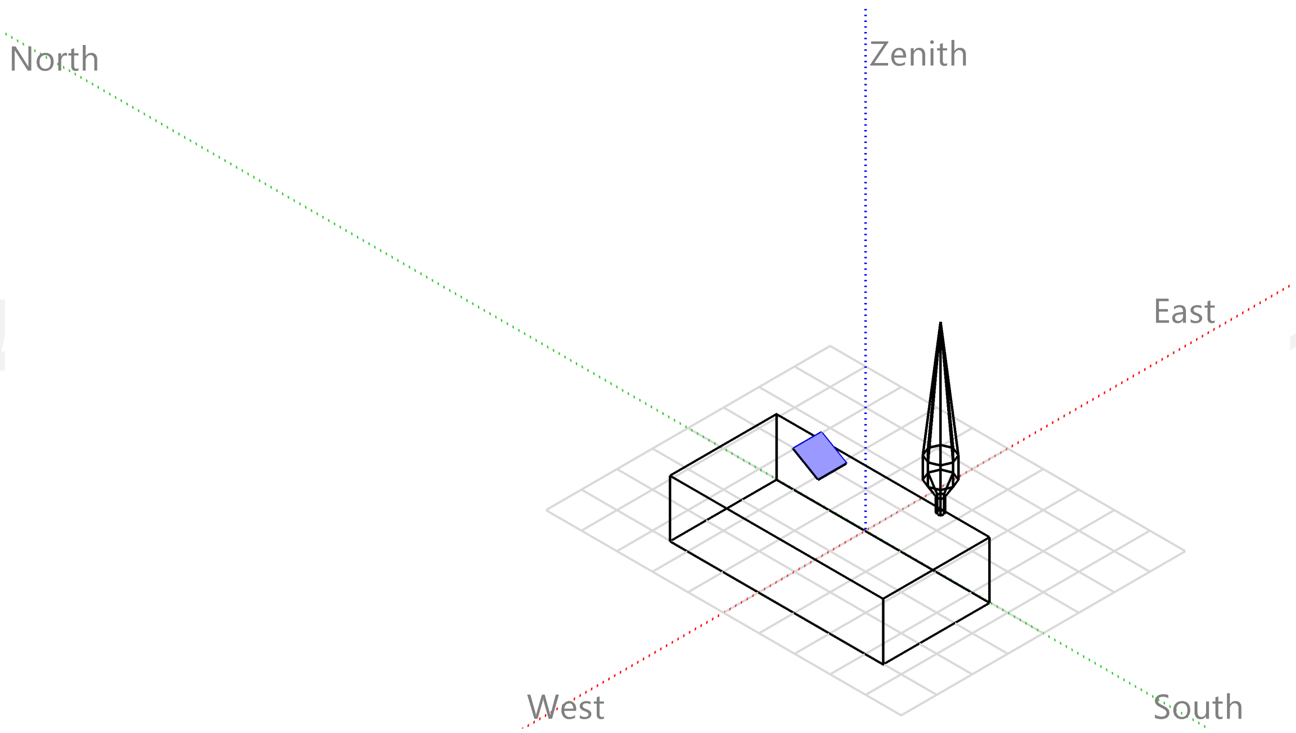
Strings Mismatch loss
Loss Fraction 0.1 %

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

Near shadings parameter

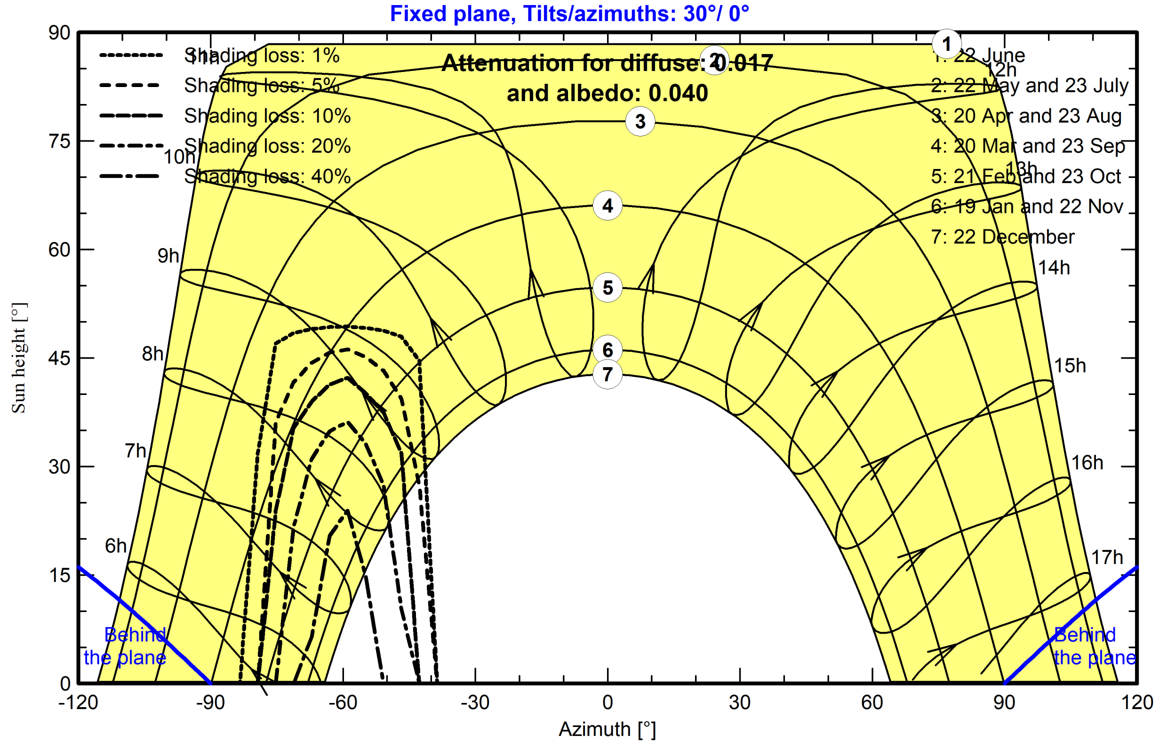
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 30°/ 0°



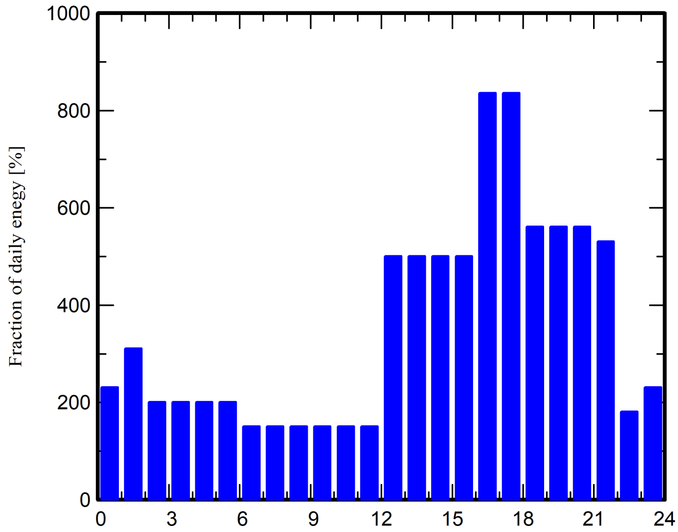
Detailed User's needs

Daily household consumers, Constant over the year, average = 8.6 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	5	12/lamp	8.0	480
TV / PC / Mobile	1	100/app	4.0	400
Domestic appliances	5	70/app	10.0	3500
Fridge / Deep-freeze	1		24	3600
Dish- and Cloth-washer	1		1	550
Stand-by consumers			24.0	24
Total daily energy				8554

Hourly distribution

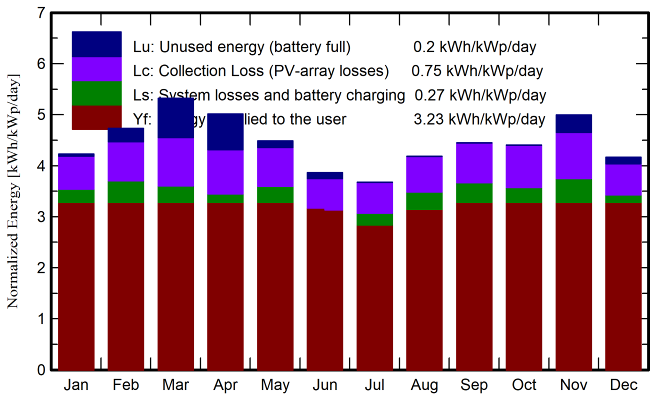


Main results

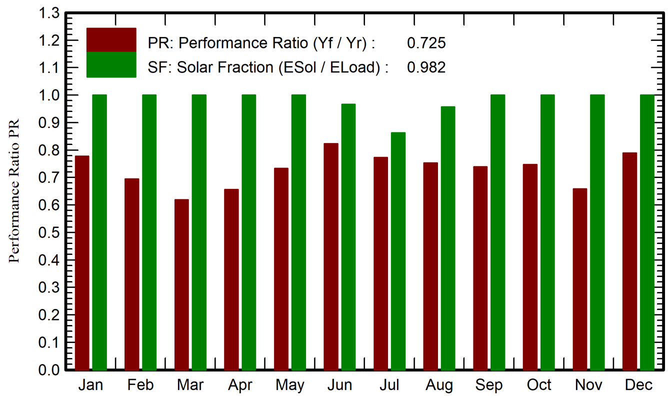
System Production

Useful energy from solar	3065.74 kWh/year	Perf. Ratio PR	72.48 %
Available solar energy	3365.08 kWh/year	Solar Fraction SF	98.19 %
Excess (unused)	193.34 kWh/year		
Loss of Load		**Battery aging (State of Wear)**	
Time Fraction	0.0 %	Cycles SOW	97.5 %
Missing Energy	56.47 kWh/year	Static SOW	90.0 %

Normalized productions (per installed kWp)



Performance Ratio PR



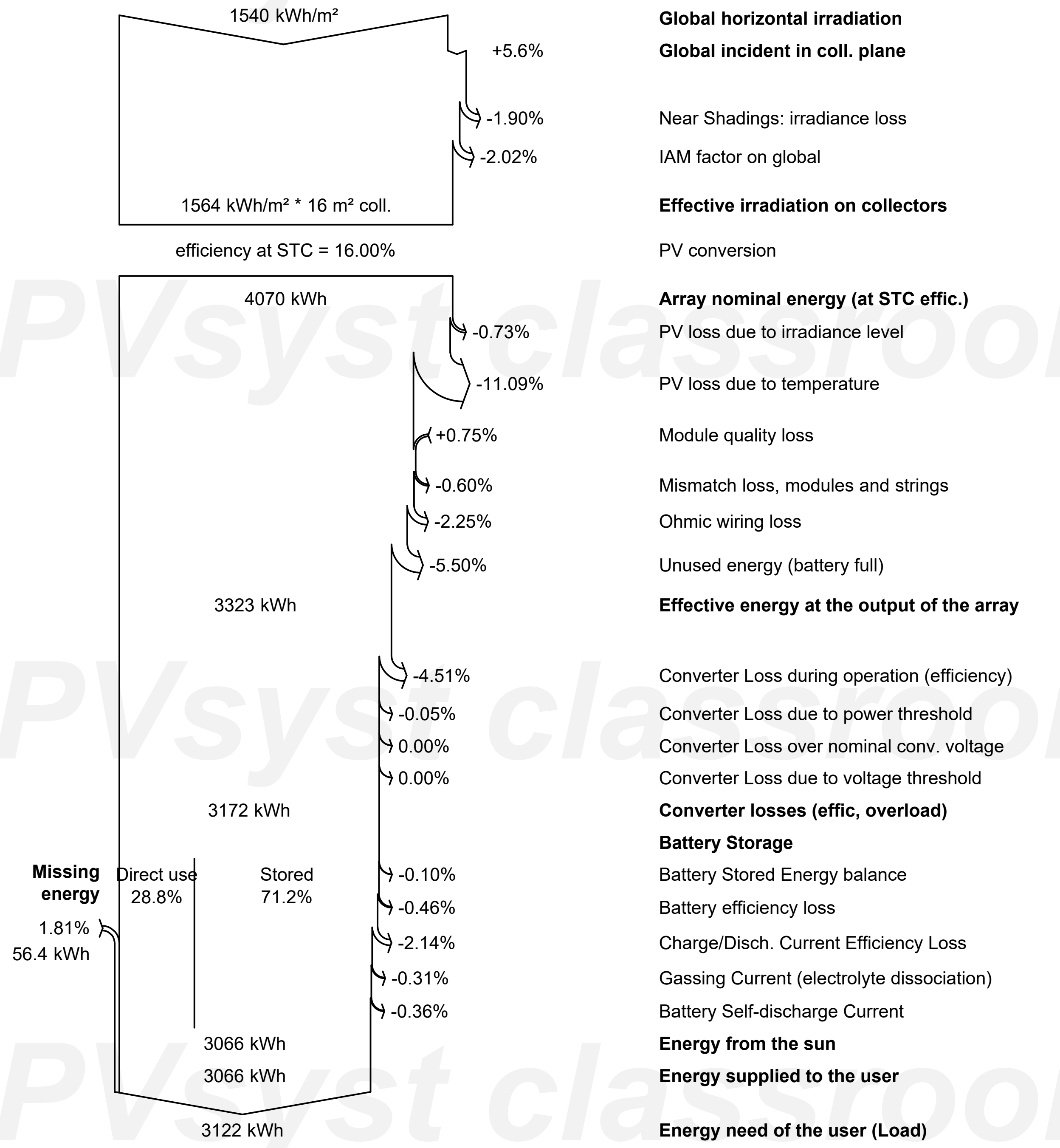
Balances and main results

	GlobHor	GlobEff	E_Avail	EUnused	E_Miss	E_User	E_Load	SolFrac
	kWh/m²	kWh/m²	kWh	kWh	kWh	kWh	kWh	ratio
January	101.8	125.1	276.0	3.13	0.00	265.2	265.2	1.000
February	113.5	126.7	277.7	19.15	0.00	239.5	239.5	1.000
March	153.0	159.4	340.2	61.46	0.00	265.2	265.2	1.000
April	153.9	146.0	311.4	54.14	0.00	256.6	256.6	1.000
May	153.4	134.5	286.6	10.14	0.00	265.2	265.2	1.000
June	131.3	111.7	240.5	8.59	8.78	247.8	256.6	0.966
July	128.5	109.8	234.0	0.01	36.34	228.8	265.2	0.863
August	138.2	125.6	267.7	0.01	11.34	253.8	265.2	0.957
September	130.5	129.0	273.4	0.00	0.00	256.6	256.6	1.000
October	120.4	129.7	275.3	0.01	0.00	265.2	265.2	1.000
November	115.5	142.4	306.9	26.46	0.00	256.6	256.6	1.000
December	99.9	124.0	275.4	10.23	0.00	265.2	265.2	1.000
Year	1540.0	1563.7	3365.1	193.34	56.47	3065.7	3122.2	0.982

Legends

GlobHor	Global horizontal irradiation	E_User	Energy supplied to the user
GlobEff	Effective Global, corr. for IAM and shadings	E_Load	Energy need of the user (Load)
E_Avail	Available Solar Energy	SolFrac	Solar fraction (EUsed / ELoad)
EUnused	Unused energy (battery full)		
E_Miss	Missing energy		

Loss diagram



Predef. graphs

Daily Input/Output diagram